

Notes and Solutions to Analysis I, Terence Tao

FRORDO

My solutions to Terence Tao's Analysis I. Enjoy!

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1. Why Analysis?

$$\lim_{x \rightarrow \infty} \sin(x) = \lim_{x \rightarrow \infty} \sin(\pi + x) = \lim_{x \rightarrow \infty} -\sin(x)$$

$$\lim_{x \rightarrow \infty} \sin x = 0$$

$$\lim_{x \rightarrow \infty} \cos x \stackrel{x \mapsto \frac{\pi}{2} - x}{=} 0$$

$$\lim_{x \rightarrow \infty} 1 = \lim_{x \rightarrow \infty} \sin^2(x) + \cos^2(x) = 0$$

$1 = 0$

2. The Natural Numbers

2.1. The Peano Axioms

Definitions

1. $\mathbb{N}$ denotes the set of natural numbers, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$
2. S is the successor function
3. Call $S(0) = 1$, $S(1) = 2$, and so on.

Axioms

1. **Axiom 1:** 0 is a natural number.
2. **Axiom 2:** If n is a natural number, $S(n)$ is a natural number
3. **Axiom 3:** 0 is not the successor of any natural number.
4. **Axiom 4:** S is a one-to-one function, that is, $S(x) = S(y) \Leftrightarrow x = y$
5. **Axiom 5 (Mathematical Induction):** Suppose $P(n)$ is a property of a natural number n . Then if $P(0)$ is true, and $P(n) \Rightarrow P(n+1)$, then P is true for all natural numbers.

Problems

1. **Axiom 1:** 0 is a natural number.
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